

GOVERNMENT OF INDIA	TECHNICAL EXPERT CERTIFICATE (PART B)
BharatShield National Deepfake Detection and Response Platform	Reference No. : EP-test-pac
RESTRICTED – LAW ENFORCEMENT USE ONLY	

Technical Expert Certificate (Part B)

Schedule to Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 · Part B

1. TECHNICAL EXPERT DETAILS

Expert Name & Designation	Dr. Authorized Forensic Examiner
Expert ID / License No.	IT-79A-IND-2026-XXXX
Verification Workstation Serial	DEMO-WS-2026-0001
Verification MAC Address	00:11:22:33:44:55
Date & Time of Examination (IST)	04-06-2026 20:54:16 IST

2. FORENSIC ANALYSIS & DETECTION METRICS

Spatial CNN Manipulation Prob.	92.0000%
3D Face Mesh Landmark Variance	0.075000
Lip-sync Alignment Error	135.00 ms
TTS Synthetic Voice Probability	88.0000%
Spectrogram Pitch Mismatch Ratio	39.0000%
Anti-spoofing NN Confidence (Bona Fide)	11.0000%

RESOLVED SUBJECT BIOMETRICS & IDENTITY PROFILE

Match Status	RESOLVED / BIOMETRIC MATCH DETECTED
Resolved Subject Name	Shri Arvind Patil
Aadhaar Number (Masked)	XXXX-XXXX-7890
Gender	Female
Electoral Role	active_candidate
Constituency	Constituency 47
Party Affiliation	National Democratic Alliance

Active Candidacy (MCC)	Yes
Face Cosine Similarity	100.0000%
Voice Cosine Similarity	100.0000%
Fused Confidence Score	100.0000%
Verification Source	hybrid

3. GROUNDS FOR DEEPPFAKE CLASSIFICATION

Summary: Media classified as likely synthetically manipulated based on 12 forensic indicator(s), including 4 high-severity finding(s).

- **Audio-Video Sync (CRITICAL):** Audio–video lip movement is misaligned, consistent with synthetic dubbing or post-hoc voice replacement rather than naturally captured speech. (lip_sync_alignment_error_ms: 135.0 ms)
- **Facial Manipulation (HIGH):** Spatial facial inconsistencies detected across frames suggest GAN or deepfake-based facial manipulation. (spatial_cnn_manipulation_probability: 92.00%)
- **3D Face Geometry (MEDIUM):** 3D face mesh landmarks are unstable across the temporal sequence, indicating inconsistent facial geometry typical of synthetic faces. (face_mesh_landmark_variance: 0.0750)
- **Synthetic Voice (HIGH):** The voice track exhibits characteristics consistent with text-to-speech or neural voice cloning rather than a live recording. (tts_synthetic_probability: 88.00%)
- **Voice Authenticity (MEDIUM):** Pitch contour and spectrogram features do not match natural speech patterns for the apparent speaker, suggesting audio synthesis or splicing. (spectrogram_pitch_mismatch_ratio: 39.00%)
- **Anti-Spoofing (HIGH):** Anti-spoofing neural network assigns low confidence to bona fide speech, indicating the audio is likely non-genuine or heavily processed. (anti_spoofing_nn_confidence: 11.00%)
- **Lighting Consistency (MEDIUM):** Lighting on the face and background shifts unnaturally between frames, suggesting compositing or generative synthesis rather than a single capture. (illumination_inconsistency_score: 71.00%)
- **Shadow Geometry (MEDIUM):** Shadow direction and softness are inconsistent with the scene light source, a common artifact in deepfake and face-swap videos. (shadow_geometry_inconsistency_score: 66.00%)
- **Color Grading (MEDIUM):** Skin tone and color grading do not remain consistent across frames; color boundaries and shadows appear imperfect relative to natural video. (color_grading_anomaly_score: 59.00%)
- **Temporal Stability (MEDIUM):** High-frequency flicker and temporal instability detected at facial boundaries, consistent with frame-wise generative manipulation. (temporal_flicker_score: 55.00%)
- **Facial Boundaries (MEDIUM):** Visible artifacts at the jawline, hairline, or face perimeter suggest blending of a synthetic face onto an original body. (facial_boundary_artifact_score: 62.00%)
- **Analyst Observation (MEDIUM):** Suspect video shows mismatched lip-sync and texture anomalies around face boundary.

4. DECLARATION & SWORN OPINION

I, Dr. Authorized Forensic Examiner, do hereby certify that the electronic record described above was subjected to multi-modal deepfake analysis on validated forensic models. It is my technical opinion, to a high degree of certainty, that the media exhibits synthetic artifacts characteristic of generative artificial intelligence impersonation. This certificate is issued under the authority of Section 79A of the Information Technology Act,

2000 for production as electronic evidence.

Dr. Authorized Forensic Examiner Digital Forensic Examiner (Sec. 79A IT Act) Forensic Analysis Centre, India	Signature: _____ Date: 04 June 2026
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